

Tab 1

HoneyPot Farm & ELK Integration

Overview

This project is based on a network of decoy systems (honeypot) to capture real time attacker data and forward that data to an ELK(Elastisearch , Logstash , Kibana) to visualize attacker behavior.

It is similar with **Deception Technology**, a core components modern, higher-maturity

SOCs. It shifts focus from just **watching alerts** to **actively hunting and analyzing** attacker **TTPs** (Tactics, Techniques, and Procedures).

Technical Stack

- **Honeypot Engine: Cowrie** (SSH/Telnet honeypot) or **Dionaea** (captures malware).
Nozomi Networks
- **Infrastructure:** Docker containers (for easy cleanup and isolation).
- **Data Pipeline: Filebeat** (to collect logs) -> **Logstash** (to parse logs) -> **Elasticsearch** (to store data).
- **Visualization: Kibana** (to create threat maps and command dashboards).

Phase I :

Note : If you don't have key then first generate the key by following command :

Command : `gpg --full-generate-key`

After generating key run the following command :

Command : `pass init <your-generated-key>`

Containerized Deployment

Deploying Cowrie by using Docker on ubuntu :

Command : `docker pull cowrie/cowrie`

```
ook@ook-Latitude-5400:~/Downloads$ docker pull cowrie/cowrie
Using default tag: latest
latest: Pulling from cowrie/cowrie
99515e7b4d35: Pulling fs layer
2780920e5dbf: Pulling fs layer
ff8b8b763ee5: Pulling fs layer
cac2ae0193cb: Pulling fs layer
47de5dd0b812: Pulling fs layer
d6b1b89eccac: Pulling fs layer
```

To use the Cowrie in Interactive mode run the following command :

Command : `docker run -d -p 8000:8000 -p 9000:9000 \`
`-v /var/run/docker.sock:/var/run/docker.sock \`

portainer/portainer-ce

docker run — Creates and starts a new container

-d — Run in detached mode (background). The container runs and you get your terminal back immediately instead of seeing live output.

-p 8000:8000 — Publish port 8000. Maps port 8000 on your host machine to port 8000 inside the container. You'll access Portainer's edge agent on <http://localhost:8000>.

-p 9000:9000 — Publish port 9000. Maps port 9000 on your host to port 9000 in the container. This is the main Portainer web UI you'll use at <http://localhost:9000>.

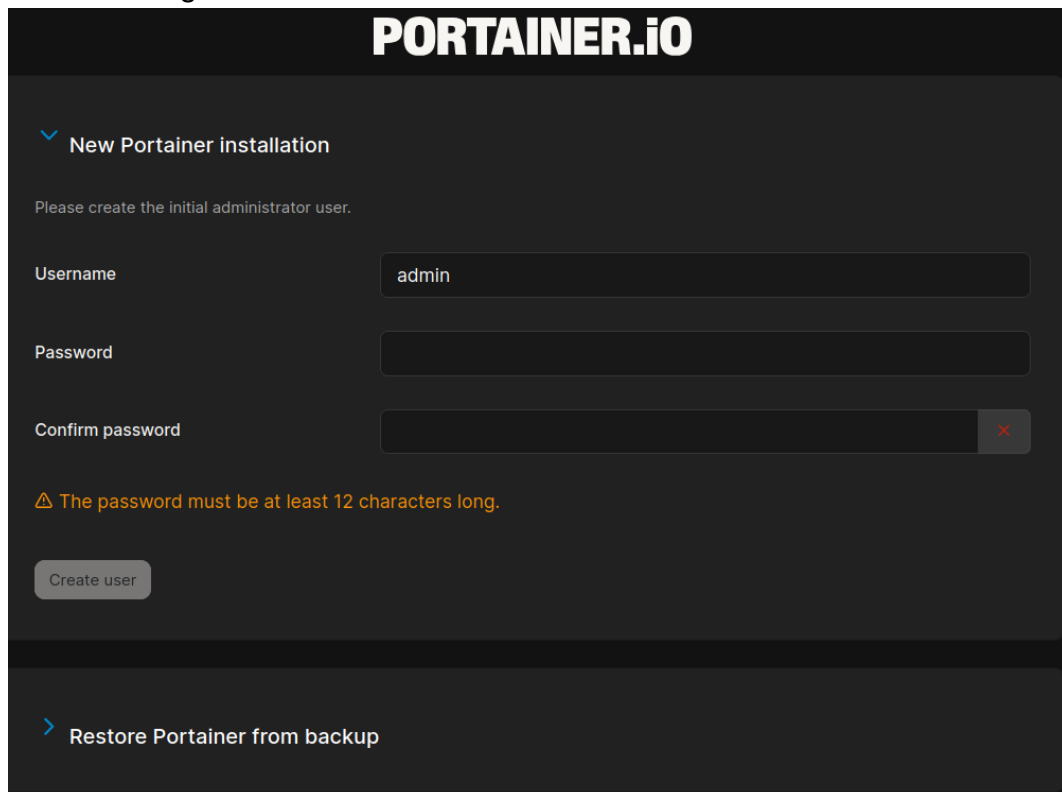
-v /var/run/docker.sock:/var/run/docker.sock — Mount a volume. This binds the Docker daemon socket from your host machine (/var/run/docker.sock) into the container at the same path. This lets Portainer communicate with Docker on your host—essentially giving the container permission to control Docker. Without this, Portainer couldn't manage your containers.

portainer/portainer-ce — The image to run. portainer-ce is the Community Edition (free, open-source).

After running this command run navigate to the web browser and enter :

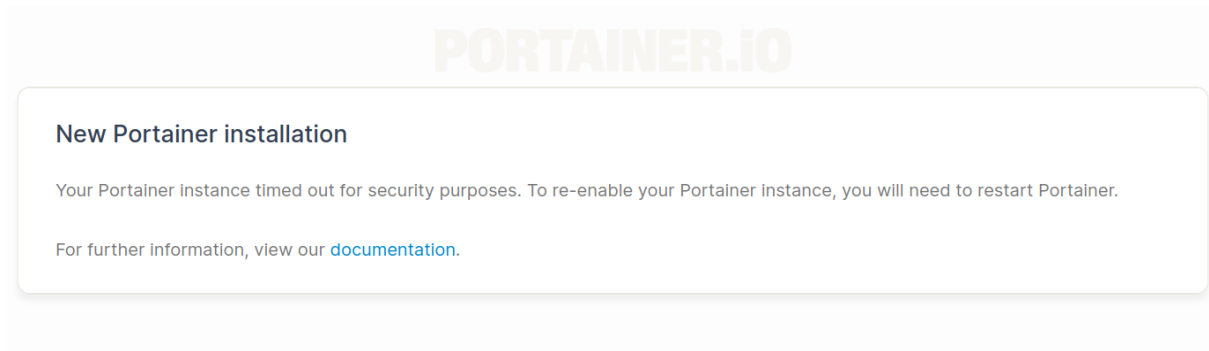
URL : <http://localhost:9000>

The following interface comes :

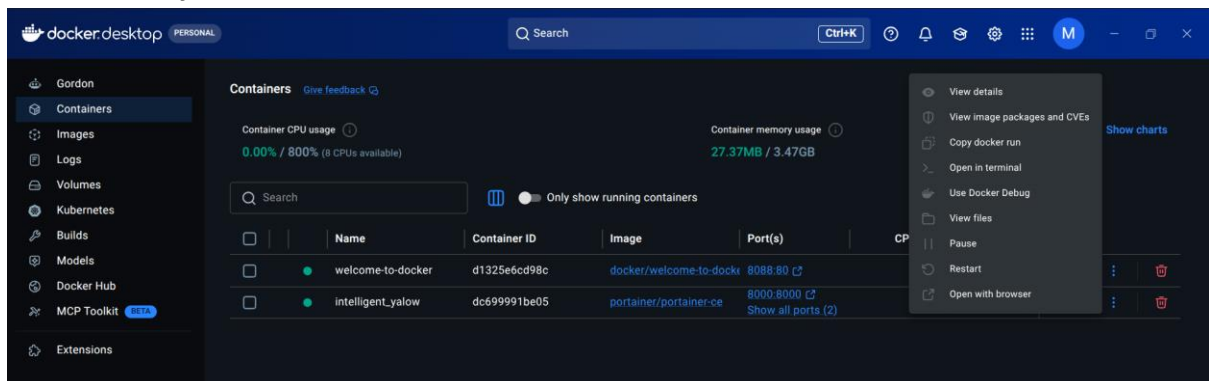


The screenshot shows the Portainer.io web interface. At the top, the logo 'PORTAINER.IO' is displayed in white on a dark background. Below the logo, there is a section titled 'New Portainer installation' with a blue checkmark icon. Under this title, a message says 'Please create the initial administrator user.' There are three input fields: 'Username' with the value 'admin', 'Password', and 'Confirm password'. The 'Confirm password' field has a red 'x' icon on its right side. Below the input fields, a warning message in orange text says '⚠ The password must be at least 12 characters long.' At the bottom of this section is a 'Create user' button. Below this section, there is another section titled 'Restore Portainer from backup' with a blue arrow icon.

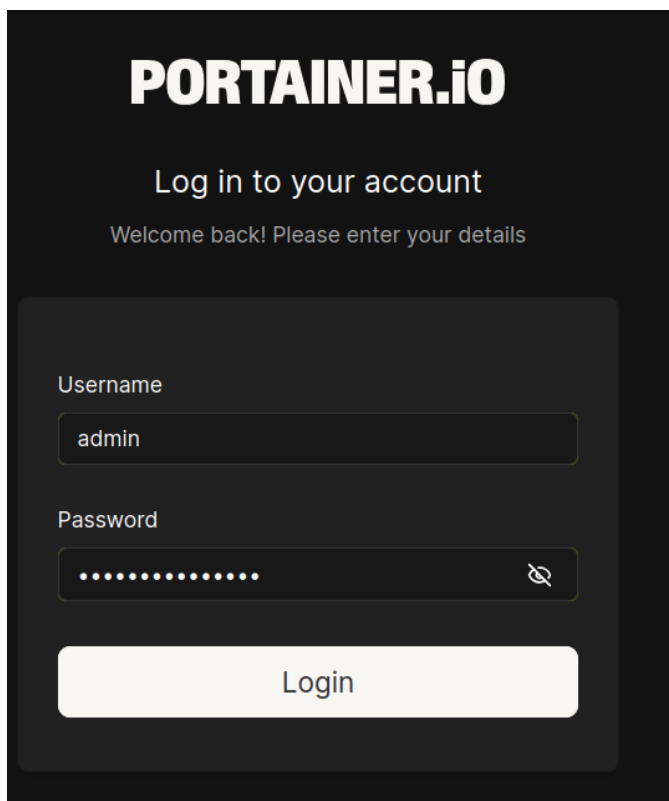
Set your password and Press **Create user**.



Now restart your **Portainer**



Click on restart. After restart it go back to the browser and the following interface shown to enter **Username** and **Password** :

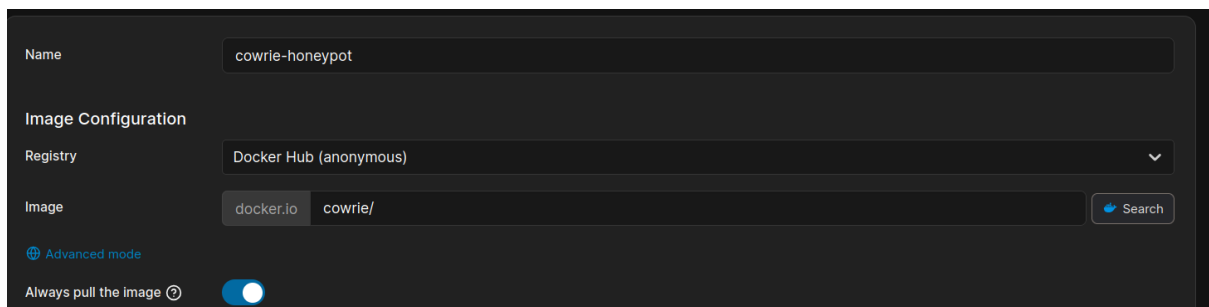


And press **login**.

After login it directs you to the **Home** page.

Deploying Cowrie via Portainer

1. Navigate to the "Containers" Section
 - On the left-hand sidebar in your Portainer dashboard, click on **Containers**.
 - Click the **+ Add container** button.
2. Configure the Container
 - **Name:** Give it a clear name, for example: `cowrie-honeypot`.
 - **Image:** In the Image field, type: `cowrie/cowrie:latest`.
 - *Note:* Portainer will automatically pull this image from Docker Hub when you click "Deploy."

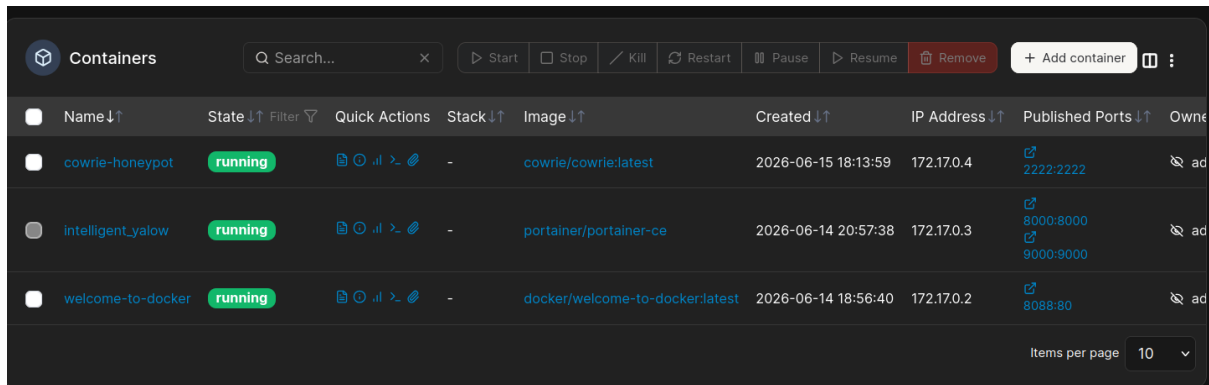
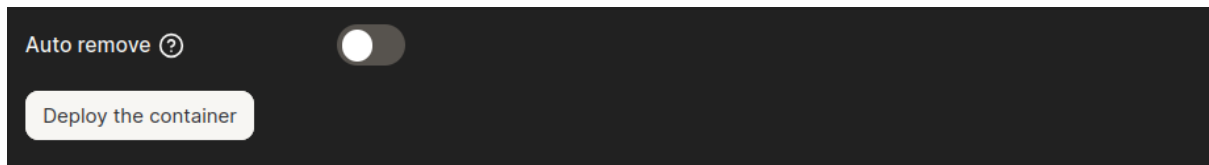


The screenshot shows the 'Add Container' form in Portainer. The 'Name' field is set to 'cowrie-honeypot'. Under 'Image Configuration', the 'Registry' is set to 'Docker Hub (anonymous)'. The 'Image' field is split into 'docker.io' and 'cowrie/'. There is a 'Search' button to the right of the image field. At the bottom, there is a link for 'Advanced mode' and a toggle for 'Always pull the image' which is currently turned on.

3. Map Network Ports
 - Scroll down to the **Network ports configuration** section.
 - Click **+ publish a new network port**.
 - **Host:** `2222`
 - **Container:** `2222`

This maps the container's internal SSH port (2222) to your host machine's port (2222). When you (or an attacker) try to SSH into your machine on port 2222, it will land inside your Cowrie honeypot.

4. Set Restart Policy (Reliability)
 - Under the **Advanced container settings**
 - Select **Always**. This ensures that if the container crashes or the system reboots, the honeypot starts back up automatically.
5. Access Container Settings :
 - Click **+ map additional volume**.
 - **Container:** Type `/cowrie/cowrie-git/var/log/cowrie/`
 - **Host:** Type `/home/"your-username"/.cowrie-logs`
6. **Deploy**
 - Click the **Deploy the container** button at the bottom of the page.
 - The portainer will now download the image and start the container. This may take a minute or two.



The Status ensures that the **cowrie-honeypot** is running. Now that the honeypot is live.

Phase II :

Verifying :

First of all create a directory which already have mentioned for **host** (i.e., **/home/"your-username"/.cowrie-logs**)

Command : `mkdir 777 -p /home/"your-username"/.cowrie-logs`

Create a fake login :

Try this command for fake login :

Command : `ssh root@127.0.0.1 -p 2222`

```
ook@ook-Latitude-5400:~/Downloads$ ssh root@127.0.0.1 -p 2222
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:zzMZN0tvcbRBDd0VQIAXfGLmuiqEY5+iEMACWEJ6X9Q.
Please contact your system administrator.
Add correct host key in /home/ook/.ssh/known_hosts to get rid of this message.
Offending ED25519 key in /home/ook/.ssh/known_hosts:1
  remove with:
  ssh-keygen -f '/home/ook/.ssh/known_hosts' -R '[127.0.0.1]:2222'
Host key for [127.0.0.1]:2222 has changed and you have requested strict checking.
Host key verification failed.
ook@ook-Latitude-5400:~/Downloads$ |
```

If you are seeing this prompt it is good news your system thinks that is a **MITM** and stops the connection immediately.

So what you need to do run the following command :

Command : `ssh-keygen -f '/home/ook/.ssh/known_hosts' -R '[127.0.0.1]:2222'`

Now enter the fake login command again :

Command : `ssh root@127.0.0.1 -p 2222`

```
ook@ook-Latitude-5400:~/Downloads$ ssh root@127.0.0.1 -p 2222
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:zzMZN0tvcbRBDd0VQIAXfGLmuiqEY5+iEMACWEJ6X9Q
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Type : **yes** and enter the fake password.

Once you try to log in, Cowrie will instantly create the log files. Type `exit` to disconnect, and then check the hidden folder one more time:

Command : `ls -la /home/ook/.cowrie-logs`

```
ook@ook-Latitude-5400:~/Downloads$ ls -la /home/ook/.cowrie-logs/
total 8
drwxrwxrwx  2 ook ook 4096 Jun 15 18:14 
drwxr-x--- 20 ook ook 4096 Jun 15 17:38 ..
-r-----  1 ook ook   0 Jun 15 18:14 cowrie.json
ook@ook-Latitude-5400:~/Downloads$ |
```

Look at the permissions the **cowrie.json** file has. It has only read only permission we need to change these permissions write the following command :

Command : `sudo chmod 666 /home/ook/.cowrie-logs/cowrie.json`

Log Pipeline Setup (Filebeat & Elasticsearch Integration)

Because we are creating this project on ubuntu. We need to install & configure **Filebeat** on the ubuntu machine.

Step 1: Install Filebeat on Ubuntu

Run these commands in your Ubuntu terminal to download and install Filebeat:

```
Command : wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg
--dearmor -o /usr/share/keyrings/elastic-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg]
https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee
/etc/apt/sources.list.d/elastic-8.x.list
sudo apt-get update && sudo apt-get install filebeat
```

```
ook@ook-Latitude-5400:~/Downloads$ wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o /usr/share/keyrings/elastic-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elastic-8.x.list
sudo apt-get update && sudo apt-get install filebeat
Hit:1 https://repo.protonvpn.com/debian stable InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:4 https://download.docker.com/linux/ubuntu resolute InRelease
Hit:5 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease
Get:6 https://artifacts.elastic.co/packages/8.x/apt stable InRelease [3,249 B]
Get:7 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 Packages [107 kB]
Hit:8 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:9 https://artifacts.elastic.co/packages/8.x/apt stable/main i386 Packages [3,396 B]
Fetched 113 kB in 2s (65.1 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following NEW packages will be installed:
  filebeat
```

Step 2 Enable Filebeat to Boot on Startup

To enable the filebeat use the following command :

Command : `sudo systemctl enable filebeat`

Step 3 Configuring Filebeat

1. Open the Filebeat Configuration File

Since this is a system configuration file, we need administrator privileges to edit it. Open it using the **nano** text editor by running this command :

Command : `sudo nano /etc/filebeat/filebeat.yml`

2. Configure the log file Input Path

Find the **filebeat.input** section. We need to change the filebeat path and direct it to our hidden file (**.cowrie-logs**)

```
# ===== Filebeat inputs =====

filebeat.inputs:

# Each - is an input. Most options can be set at the input level, so
# you can use different inputs for various configurations.
# Below are the input-specific configurations.

# filestream is an input for collecting log messages from files.
- type: filestream

  # Unique ID among all inputs, an ID is required.
  id: my-filestream-id

  # Change to true to enable this input configuration.
  enabled: false

  # Paths that should be crawled and fetched. Glob based paths.
  paths:
    - /var/log/*.log
    #- c:\programdata\elasticsearch\logs\*
```

Modern versions of Filebeat use **filestream** as the default input type rather than the older **log** type, which is perfectly fine.

```
# ===== Filebeat inputs =====

filebeat.inputs:

# Each - is an input. Most options can be set at the input level, so
# you can use different inputs for various configurations.
# Below are the input-specific configurations.

# filestream is an input for collecting log messages from files.
- type: filestream

  # Unique ID among all inputs, an ID is required.
  id: cowrie-honeypot-logs

  # Change to true to enable this input configuration.
  enabled: true

  # Paths that should be crawled and fetched. Glob based paths.
  paths:
    - /home/ook/.cowrie-logs/cowrie.json
    #- c:\programdata\elasticsearch\logs\*
  parsers:
    - ndjson:
        keys_under_root: true
        overwrite_keys: true
```

then save it (**CTRL+O, Enter, CTRL+X**).

3. Restart Filebeat

Restart the filebeat to apply the changes that we made in it.

Command : `sudo systemctl restart filebeat`

If the terminal empty returns then it means changes are successfully applied.

4. Downloading Elasticsearch

Run the following command to Download elasticsearch

Command : `sudo apt install elasticsearch`

```
----- Security autoconfiguration information -----
Authentication and authorization are enabled.
TLS for the transport and HTTP layers is enabled and configured.

The generated password for the elastic built-in superuser is : QZUElB8VXEDcYKzBUtr

If this node should join an existing cluster, you can reconfigure this with
'/usr/share/elasticsearch/bin/elasticsearch-reconfigure-node --enrollment-token <token-here>'
after creating an enrollment token on your existing cluster.

You can complete the following actions at any time:

Reset the password of the elastic built-in superuser with
'/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic'.

Generate an enrollment token for Kibana instances with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana'.

Generate an enrollment token for Elasticsearch nodes with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s node'.

-----
### NOT starting on installation, please execute the following statements to configure elasticsearch service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch.service
### You can start elasticsearch service by executing
sudo systemctl start elasticsearch.service
Processing triggers for procps (2:4.0.4-9ubuntu1)...
Processing triggers for systemd (259.5-0ubuntu3)...
ook@ook-Latitude-5400:~/Downloads$ |
```

Note : Save the Elasticsearch Password into your file. We will use it later!!!

After installation Completed run the following commands :

Command : `sudo systemctl daemon-reload`

`sudo systemctl enable elasticsearch.service`

`sudo systemctl start elasticsearch.service`

If you saw this kind of error during the start of elasticsearch then follow the following steps otherwise move to the next step.

- **Create an Options Modification File**

Instead of editing the core configuration files directly, it is best practice to create a small override file inside the **jvm.options.d** directory.

Open a new file using **nano**

Command : `sudo nano`

`/etc/elasticsearch/jvm.options.d/memory.options`

- Add the Memory Constraints

Type or paste the following configuration lines into the file. This tells Elasticsearch to start with 1 GB of RAM and cap its maximum consumption at 2 GB:

Past : `-Xms1g`

`-Xms2g`

Save the file (Press **Ctrl+O** Enter , **Ctrl+X**)

- **Restart , Start and Status of Elasticsearch**

Command : `sudo systemctl restart elasticsearch`

`sudo systemctl start elasticsearch`

`sudo systemctl status elasticsearch`

```
ook@ook-Latitude-5400:~/Downloads$ sudo systemctl restart elasticsearch
ook@ook-Latitude-5400:~/Downloads$ sudo systemctl start elasticsearch
ook@ook-Latitude-5400:~/Downloads$ sudo systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-15 20:30:59 PKT; 20s ago
     Invocation: 4bfc7f15cac04c6b94b261176b0d3a61
       Docs: https://www.elastic.co
    Main PID: 262775 (java)
      Tasks: 126 (limit: 18290)
     Memory: 2.6G (peak: 2.6G)
        CPU: 1min 7.077s
    CGroup: /system.slice/elasticsearch.service
            └─262775 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli.script=/usr/share/elasticsearch
               └─262851 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPrefer
                  └─262881 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller

Jun 15 20:30:34 ook-Latitude-5400 systemd[1]: Starting elasticsearch.service - Elasticsearch...
Jun 15 20:30:59 ook-Latitude-5400 systemd[1]: Started elasticsearch.service - Elasticsearch.
ook@ook-Latitude-5400:~/Downloads$
```

5. Changing the filebeat configuration file for Elasticsearch

Write the following command in the terminal :

Command : `sudo nano /etc/filebeat/filebeat.yml`

Find **Elasticsearch Output**

```
# ----- Elasticsearch Output -----
output.elasticsearch:
  # Array of hosts to connect to.
  hosts: ["localhost:9200"]

  # Performance preset - one of "balanced", "throughput", "scale",
  # "latency", or "custom".
  preset: balanced

  # Protocol - either 'http' (default) or 'https'.
  #protocol: "https"
```

Remove the **hosts** and past the following lines :

`hosts: ["https://localhost:9200"]`

`username: "elastic"`

`password: "Elasticsearch-password"`

`ssl.verification_mode: "none"`

Note : Enter the Password which I said to save after the elasticsearch installation is completed.

Always after changing restart the filebeat

Command : `sudo systemctl restart filebeat`

Test that does filebeat is working perfectly or not use this command

Command : `sudo filebeat test output`

```
ook@ook-Latitude-5400:~/Downloads$ sudo filebeat test output
elasticsearch: https://localhost:9200...
  parse url... OK
  connection...
    parse host... OK
    dns lookup... OK
    addresses: 127.0.0.1
    dial up... OK
  TLS...
    security... WARN server's certificate chain verification is disabled
    handshake... OK
    TLS version: TLSv1.3
    dial up... OK
  talk to server... OK
  version: 8.19.16
ook@ook-Latitude-5400:~/Downloads$
```

Phase III

Elasticsearch Indexing

The connection is active.

Now first do some activity run the following command :

Command : `ssh wronguser@127.0.0.1 -p 2222`

When it say for password to you then enter wrong password (**mrsonic**)

```
ook@ook-Latitude-5400:~/Desktop$ ssh wronguser@127.0.0.1 -p 2222
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
wronguser@127.0.0.1's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
wronguser@svr04:~$ ls
wronguser@svr04:~$ cd ../../
wronguser@svr04:/$ ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
wronguser@svr04:/$ exit
Connection to 127.0.0.1 closed.
```

Now we need to check whether elasticsearch has begun indexing to cowrie logs or not. Run the following command :

Command : `curl -k -u elastic -X GET "https://localhost:9200/_cat/indices?v"`

added **-k** to tell curl to ignore the self-signed certificate warning, and **-u elastic** so it prompts for password.

Type Elastic Password :

```
ook@ook-Latitude-5400:~/Downloads$ curl -k -u elastic -X GET "https://localhost:9200/_cat/indices?v"
Enter host password for user 'elastic':|
```

Now enter the elastic password which we have saved after the installation of elasticsearch is completed.

```
ook@ook-Latitude-5400:~/Desktop$ curl -k -u elastic -X GET "https://localhost:9200/_cat/indices?v"
Enter host password for user 'elastic':
health status index      uuid                                pri rep docs.count docs.deleted store.size pri.store.size dataset.size
yellow open      .ds-filebeat-8.19.16-2026.06.17-000001 xjff-5w5IRuiv0LCi8f6lwQ  1   1         15             0        64.6kb         64.6kb         64.6kb
ook@ook-Latitude-5400:~/Desktop$ |
```

Now the data is indexed, the next step of Phase III is learning to read and search specific parameters (like Attacker IPs , used passwords and typed commands) using elasticsearch queries.

Lets try to extract the password that we have typed during the honeypot test. Run the following json query in the terminal.

Command : `curl -k -u elastic -X GET "https://localhost:9200/.ds-filebeat-8.19.16-2026.06.17-000001/_search?pretty" -H 'Content-Type: application/json' -d'`

```
{
  "_source": ["eventid", "src_ip", "message"],
  "query": {
    "match": {
      "eventid": "cowrie.command.input"
    }
  }
}
```

```
root@book-Latitude-5400:~/Desktop# curl -k -u elastic -X GET "https://localhost:9200/_ds-filebeat-8.19.16-2026.06.17-000001/_search?pretty" -H 'Content-Type: application/json' -d '{
  "source": ["eventid", "src_ip", "message"],
  "query": {
    "match": {
      "eventid": "cowrie.command.input"
    }
  }
}
```

Enter host password for user 'elastic':

```
{
  "took": 12,
  "timed_out": false,
  "shards": {
    "-total": 1,
    "successful": 1,
    "skipped": 0,
    "failed": 0
  },
  "hits": {
    "total": {
      "value": 5,
      "relation": "eq"
    },
    "max_score": 1.0678406,
    "hits": [
      {
        "_index": ".ds-filebeat-8.19.16-2026.06.17-000001",
        "_id": "3E0qIp4B6jFeRmR0l_78",
        "_score": 1.0678406,
        "_source": {
          "message": "CMD: cd ../..",
          "src_ip": "192.168.65.1",
          "eventid": "cowrie.command.input"
        }
      },
      {
        "_index": ".ds-filebeat-8.19.16-2026.06.17-000001",
        "_id": "3UQqIp4B6jFeRmR0l_78",
        "_score": 1.0678406,
        "_source": {
          "message": "CMD: ls",
          "eventid": "cowrie.command.input",
          "src_ip": "192.168.65.1"
        }
      },
      {
        "_index": ".ds-filebeat-8.19.16-2026.06.17-000001",
        "_id": "3kQqIp4B6jFeRmR0l_78",
        "_score": 1.0678406,
        "_source": {
          "src_ip": "192.168.65.1",
          "message": "CMD: exit",
          "eventid": "cowrie.command.input"
        }
      }
    ]
  }
}
```

Here are some commands which are shown. It means our honeypot is ready now we just need to visualise data.

So that is why we are installing Kibana. **(Kibana is described in the kibana documentation file. So, check it out first).**

Phase IV

Kibana Installation

To install kibana by terminal run the command :

Command : `sudo apt install kibana`

```
hokbook-Latitude-5400:~/Desktop$ sudo apt install kibana
[sudo: authenticate] Password:
Installing:
  kibana

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 3
  Download size: 400 MB
  Space needed: 1,239 MB / 103 GB available

Get:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 kibana amd64 8.19.16 [400 MB]
Fetched 400 MB in 3min 45 (2.171 kB/s)
Selecting previously unselected package kibana.
(Reading database ... 182251 files and directories currently installed.)
Preparing to unpack ./kibana_8.19.16_amd64.deb...
Unpacking kibana (8.19.16)...
Setting up kibana (8.19.16)...
Creating kibana group... OK
Creating kibana user... OK
Kibana is currently running with legacy OpenSSL providers enabled! For details and instructions on how to disable see https://www.elastic.co/guide/en/kibana/8.19/production.html#openssl-legacy-provider
Created kibana.keystore in /etc/kibana/kibana.keystore.
```

Once the installation finishes , run the following command to start , enable and check the status of kibana :

Command : `sudo systemctl start kibana && sudo systemctl enable kibana && sudo systemctl status kibana`

```

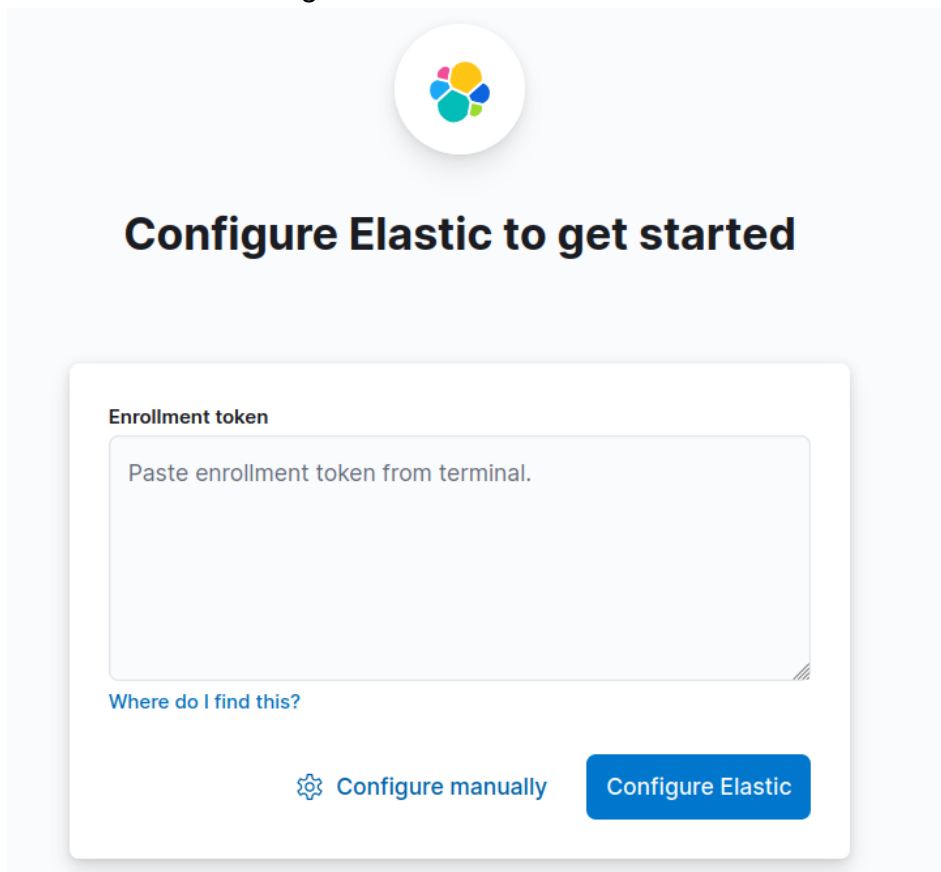
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-06-17 22:57:26 PKT; 1min 9s ago
   Invocation: ff39f43181334f679e52de2dff4fb3e5
     Docs: https://www.elastic.co
    Main PID: 43877 (node)
      Tasks: 11 (limit: 38293)
     Memory: 344.3M (peak: 524.4M)
        CPU: 9.824s
    CGroup: /system.slice/kibana.service
           └─43877 /usr/share/kibana/bin/../../node/glibc-217/bin/node /usr/share/kibana/bin/../../src/cli/kibana/dist

Jun 17 22:57:27 ook-Latitude-5400 kibana[43877]: Native global console methods have been overridden in production environment.
Jun 17 22:57:28 ook-Latitude-5400 kibana[43877]: [2026-06-17T22:57:28.182+05:00][INFO ][root] Kibana is starting
Jun 17 22:57:28 ook-Latitude-5400 kibana[43877]: [2026-06-17T22:57:28.198+05:00][INFO ][node] Kibana process configured with roles: [background_tasks, ui]
Jun 17 22:57:33 ook-Latitude-5400 kibana[43877]: [2026-06-17T22:57:33.265+05:00][INFO ][plugins-service] The following plugins are disabled: ["cloudChat, clo
Jun 17 22:57:33 ook-Latitude-5400 kibana[43877]: [2026-06-17T22:57:33.314+05:00][INFO ][http.server.Preboot] http server running at http://localhost:5601
Jun 17 22:57:33 ook-Latitude-5400 kibana[43877]: [2026-06-17T22:57:33.389+05:00][INFO ][plugins-system.preboot] Setting up [1] plugins: [interactiveSetup]
Jun 17 22:57:33 ook-Latitude-5400 kibana[43877]: [2026-06-17T22:57:33.400+05:00][INFO ][preboot] "interactiveSetup" plugin is holding setup: Validating Ela
Jun 17 22:57:33 ook-Latitude-5400 kibana[43877]: [2026-06-17T22:57:33.424+05:00][INFO ][root] Holding setup until preboot stage is completed.
Jun 17 22:57:41 ook-Latitude-5400 kibana[43877]: i Kibana has not been configured.
Jun 17 22:57:41 ook-Latitude-5400 kibana[43877]: Go to http://localhost:5601/?code=325790 to get started.
~
~

```

Open Kibana in Your Browser

Look closely at the last line of kibana status **Go to <http://localhost:5601/?code=325790> to get started.** Kibana has automatically generated links for us. Let's use it. Go to the browser and run the given link.



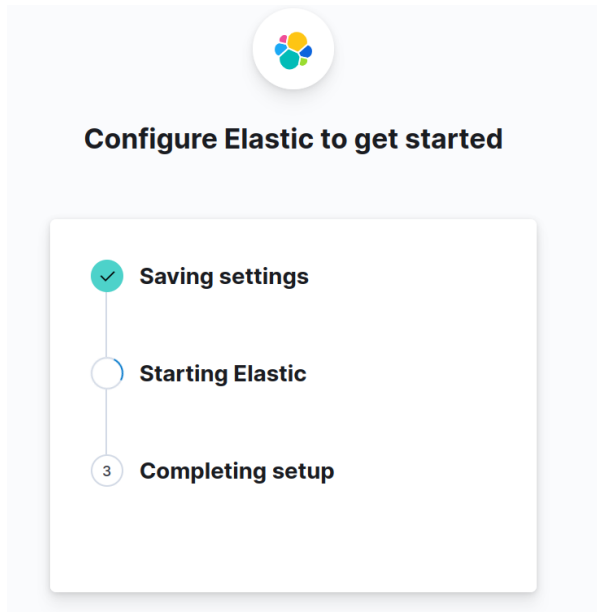
Here **kibana** is asking as for the token to configure the **Elastic** run the following command to generate the token :

Command : `sudo /usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana`

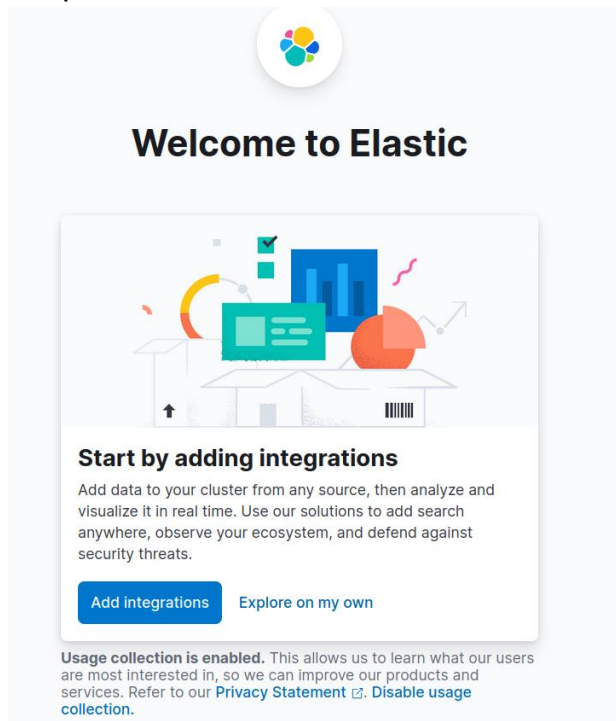
Generate an Elasticsearch Enrollment Token

Note : here is a twist when we installed the elasticsearch if you know that we also have done some security configuration by **TLSv1.3** and its session uses a cipher suite

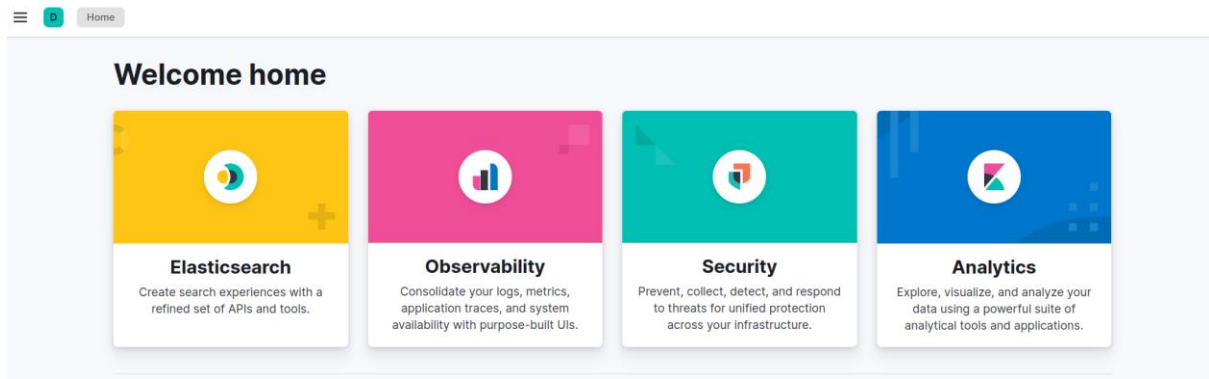
(**TLS_AES_256_GCM_SHA384**). The certificate includes several Subject Alternative Names. So, it's very important to remain yourself in the same network for this **honeypot**. If you do not then this time when you want the token for kibana to configure the elasticsearch becomes not possible. So, make sure of your **availability**.



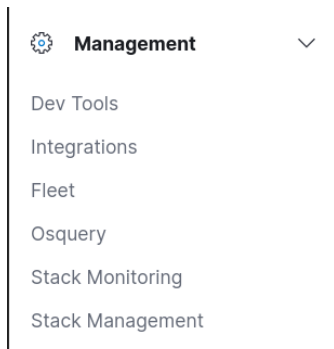
After that's completed. The next login page comes to enter your elastic username and password.



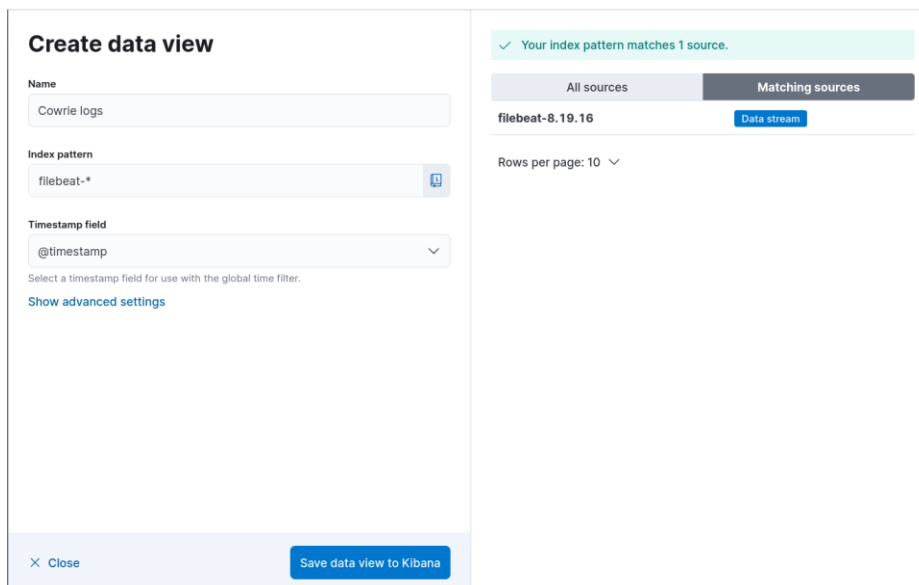
Click on **Explore on my own**
The following home page will comes :



Click on **three lines** and navigate to **Stack Management**



Then on the next page left side scroll down and find **Data Views**



Click on **create a data view**

Set name = **Cowrie logs**

Index pattern = **filebeat-*** (it can automatically match it)

Timestamp Field = **@timestamp**

Then click on **save data view to kibana**

Cowrie logs

[Delete](#)[Edit](#)Index pattern: [filebeat-*](#) Time field: [@timestamp](#) [★ Default](#)[Fields \(6682\)](#) [Scripted fields \(0\)](#) [Field filters \(0\)](#) [Relationships \(0\)](#)

Q Search

Field type16

Schema type2

Refresh

Add field

Name ↑	Type ↓	Format ↓	Searchable ↓	Aggregatable ↓	Excluded ↓	Actions
@timestamp	date		●	●		✎
_id	_id		●			✎
_ignored	_ignored		●	●		✎
_index	_index		●	●		✎
_score						✎
_source	_source					✎
activemq.caller	keyword		●	●		✎
activemq.log.stack_trace	keyword		●	●		✎
activemq.thread	keyword		●	●		✎
activemq.user	keyword		●	●		✎

Rows per page: 10

<

1

2

3

4

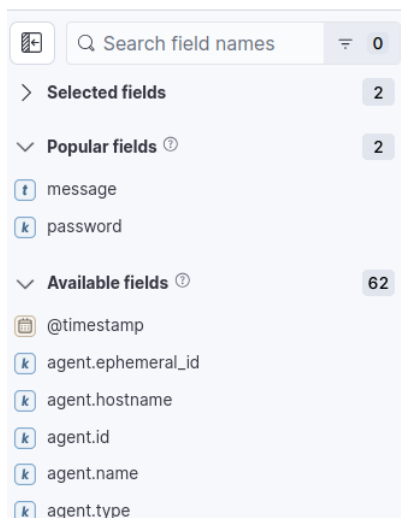
5

...

669

>

Now after that select the **Discover** option from the **top left panel**.

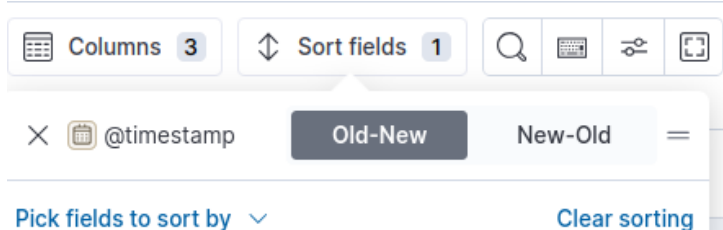


In this given search field search :

- 1) Password
- 2) Message

And drag them one by one to the main field

To arrange all the process step by step go to the **Sort field** and select **timestamp**



As shown in the screenshot

Now you can visualise every process step by step that what the hacker does step by step from last to end :

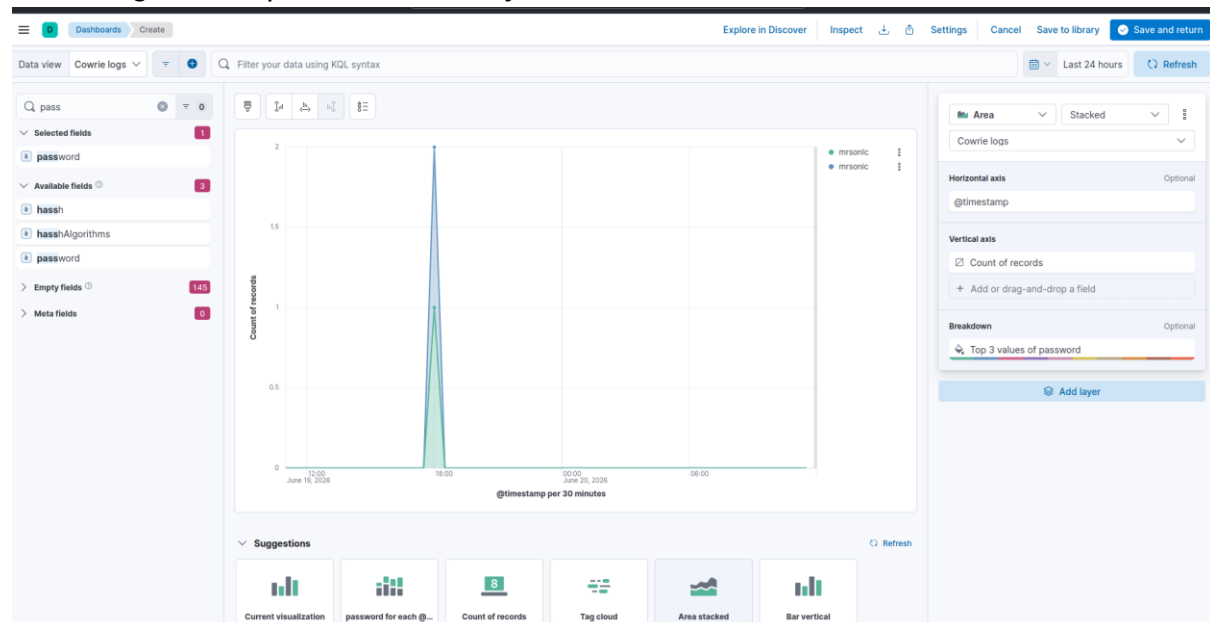
Documents (15) Field statistics			Columns 3	Sort fields 1	Search	Filter	Visualize	Export
☐	@timestamp	password	message					
☐	Jun 17, 2026 @ 20:08:17.542	-	New connection: 192.168.65.1:19266 (172.17.0.2:2222) [session: ed45e04dbaae]					
☐	Jun 17, 2026 @ 20:08:17.544	-	Remote SSH version: SSH-2.0-OpenSSH_10.2p1 Ubuntu-2ubuntu3.2					
☐	Jun 17, 2026 @ 20:08:17.544	-	SSH client hash fingerprint: eeca246859b9ded084ecf2f70a75356					
☐	Jun 17, 2026 @ 20:08:17.545	lsdkcs	Login attempt [wronguser/lsdkcs] succeeded					
☐	Jun 17, 2026 @ 20:08:17.545	-	Terminal Size: 217 51					
☐	Jun 17, 2026 @ 20:08:17.545	-	request_env: COLORTERM=truecolor					
☐	Jun 17, 2026 @ 20:08:17.545	-	request_env: LANG=en_US.UTF-8					
☐	Jun 17, 2026 @ 20:08:17.545	-	-					
☐	Jun 17, 2026 @ 20:08:19.547	-	CMD: ls					
☐	Jun 17, 2026 @ 20:08:25.551	-	CMD: pwd					
☐	Jun 17, 2026 @ 20:08:27.552	-	CMD: cd ../..					
☐	Jun 17, 2026 @ 20:08:29.553	-	CMD: ls					
☐	Jun 17, 2026 @ 20:08:31.553	-	CMD: exit					
☐	Jun 17, 2026 @ 20:08:31.553	-	Closing TTY Log: var/lib/cowrie/tty/1349e36d2e0552f1d984cc565746e16bef38e31fa7e35989326ef7b877a316b9 after 15.7 seconds					
☐	Jun 17, 2026 @ 20:08:31.554	-	Connection lost after 18.1 seconds					

After completing it.

Click on the  and find **dashboard** option :

Create a new dashboard

Now drag and drop the field which you want to select for dashboard :



You are able to change the visualisation in which style you want.

And also save it by pressing the **Save and return** button.

Good Bye!!!

